Roll No……..

MID SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program: ***B.Tech*** | *Semester:* **V**

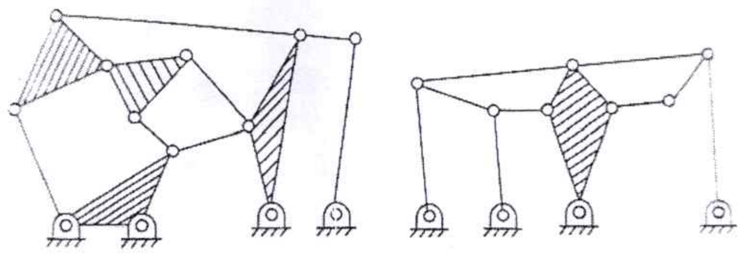
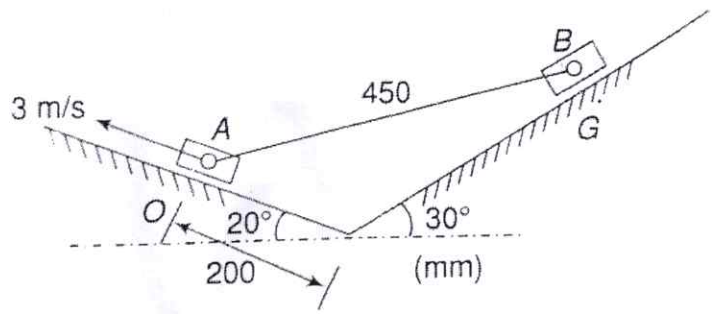
Name of the Course: **Theory of Machines** | *Course Code:* **TME 508**

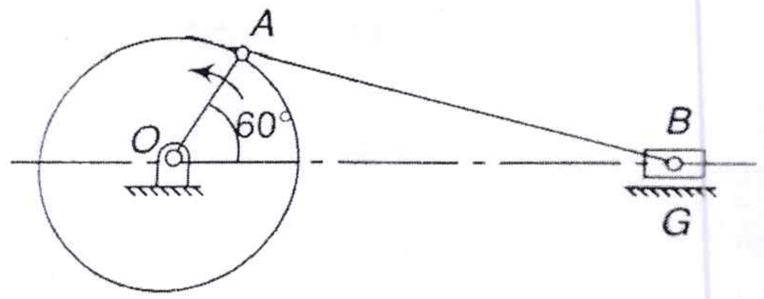
Time: 90 Minutes | Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

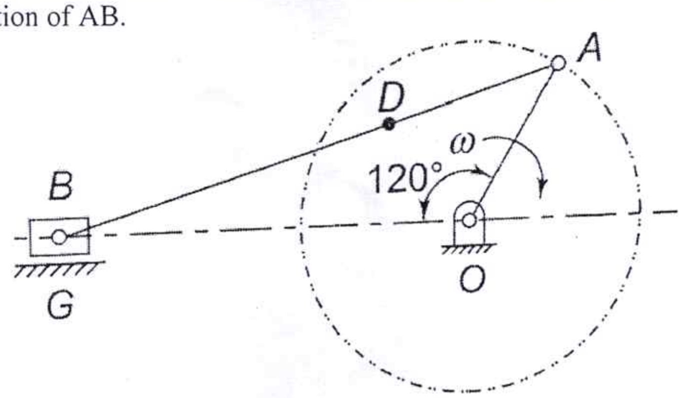
(ii) Each question carries 10 marks

Q1	(10 marks)	
(a)	How are the kinematic pairs classified? Explain with examples.	CO1
	OR	
(b)	Find Degree of freedom of the given kinematic linkages:  (a) (b)	
Q2	(10 marks)	
(a)	For the position of the mechanism shown in figure, find the velocity of the slider B for the given configuration if the velocity of the slider A is 3 m/s. 	CO2
	OR	
(b)	In a slider-crank mechanism, the crank is 480 mm long and rotates at 20 rad/s in the counter-clockwise direction The length of the connecting rod is 1.6 m. When the crank turns 60° from the inner-dead centre, determine the velocity of the slider.	



Q3 (10 marks)

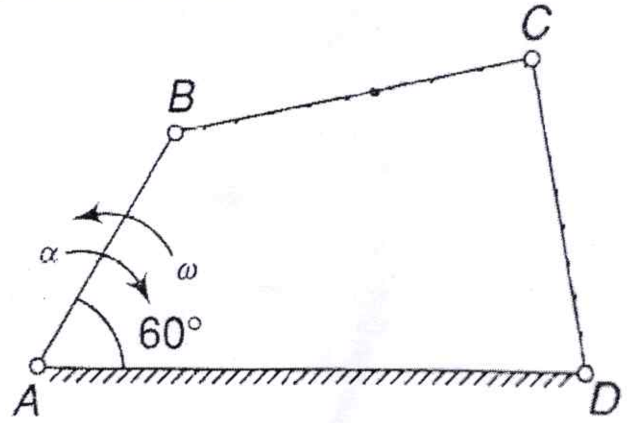
(a) Figure shows configuration of an engine mechanism. The dimensions are the following: Crank OA = 200 mm; Connecting rod AB = 600 mm, AD = 200 mm. At the instant, the crank has an angular velocity of 50 rad/s clockwise and an angular acceleration of 800 rad/s^2. Calculate acceleration of D and angular acceleration of AB.



CO2

OR

(b) Figure shows the configuration diagram of a four-link mechanism (AB=50mm, BC=66mm, CD=56mm, AD=100mm). The link AB has an instantaneous angular velocity of 10.5 rad/s and a retardation of 26 rad/s^2. Find the angular accelerations of the links BC and CD.



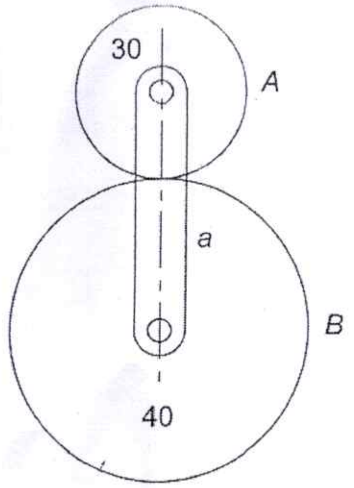
Q4 (10 marks)

(a) Define the terms:

i. Pitch circle
ii. Pitch diameter
iii. Pitch point
iv. Circular pitch
v. Module

CO3

OR

(b)	State and derive the law of gearing.	
Q5	(10 marks)	
(a)	An epicyclic gear train consists of an arm and two gears A and B having 30 and 40 teeth respectively. The arm rotates about the centre of the gear A at a speed of 80 rpm counter-clockwise. Determine the speed of the gear B if: i. The gear A is fixed ii. The gear A revolves at 240 rpm clockwise instead of being fixed.  	**CO3**
	OR	
(b)	An epicyclic gear train is shown in figure. The number of teeth on A and B are 80 and 200. Determine the speed of the arm a: i. If A rotates at 100 rpm clockwise and B at 50 rpm counter-clockwise ii. If A rotates at 100 rpm clockwise and B is stationary